



CYBERSECURITY DEFENSE OPERATIONS // SUITE v1.4.2

PHISHGUARD

Next-Generation Tactical Phishing Defense, Interactive Threat Lab & Security
Operations Center (SOC) Platform

• 10 CORE OPERATIONAL MODULES

ANDROID APK + iOS PWA + WEB

ZERO TELEMETRY // PRIVACY FIRST

Live Public URL: <https://shlokt05.github.io/phishguard/> | Production Build: Oct 2026



The Modern Threat Landscape & Mission

Why traditional antivirus and passive training fail against weaponized social engineering.

3.4B+

PHISHING EMAILS SENT DAILY

91%

CYBERATTACKS BEGIN WITH PHISHING

100%

HANDS-ON INTERACTIVE DEFENSE IN PHISHGUARD

⚠ Critical Cybersecurity Vulnerabilities

- ▶ **Weaponized AI Clones:** Attackers create pixel-perfect clones of banking and corporate login portals in minutes.
- ▶ **Smishing & Quishing Surges:** Malicious SMS messages and QR codes bypass standard email spam filters.
- ▶ **Psychological Manipulation:** Fabricated urgency ("Account Blocked", "Immediate KYC") triggers impulsive human errors.
- ▶ **Passive Training Failure:** Reading static articles does not train human muscle memory for real attacks.

🛡 The PhishGuard Strategic Solution

- ▶ **Interactive Threat Lab:** Side-by-side inspection of live clones with radar-guided forensic hotspots.
- ▶ **Dual Heuristics Engine:** Real-time risk analysis for suspicious URLs and deceptive SMS text.
- ▶ **SOC Readiness Posture:** Real-time readiness meter tracking threat identification competency.
- ▶ **Universal Accessibility:** Native Android APK, zero-friction iOS Safari PWA, and public web access.

End-to-End Engineering Architecture

A high-performance, modular stack engineered for zero latency, offline readiness, and universal deployment.

TIER 1: PRESENTATION

Frontend Engine

- ▶ **React 18 + TypeScript:** Type-safe state and reactive component trees.
- ▶ **Vite 5 Build Tool:** Sub-second HMR and optimized production bundling.
- ▶ **Tailwind CSS:** Anti-AI Dark Tactical palette (#0A0E17 slate).
- ▶ **Lucide Icons:** Precision vector iconography.

TIER 2: CROSS-PLATFORM

Mobile & PWA Layer

- ▶ **Capacitor 6 Runtime:** Native Android bridge and asset syncing.
- ▶ **Full Mipmap Matrix:** mdpi, hdpi, xdpi, xxhdpi, xxxhdpi app icons.
- ▶ **iOS Safari PWA:** Web App Manifest, offline caching, and Add-to-Home.
- ▶ **HashRouter:** 100% route immunity against 404s on static CDNs.

TIER 3: HEURISTICS

Threat Detection

- ▶ **Entropy Analysis:** Shannon entropy checks for obfuscated domains.
- ▶ **TLD Reputation:** Heuristic filters for high-risk TLDs (.top, .buzz).
- ▶ **Smishing NLP Engine:** Panic keyword & fraud trigger detection.
- ▶ **Client-Side Execution:** Zero network roundtrip; privacy guaranteed.

TIER 4: DEVOPS & CI/CD

Global Edge Delivery

- ▶ **GitHub Actions CI/CD:** Automatic build and deploy on git push.
- ▶ **GitHub Pages CDN:** Global Fastly edge servers with TLS 1.3.
- ▶ **Static Fallback:** Dual 200.html & 404.html single-page handlers.
- ▶ **24/7 Availability:** Zero server maintenance overhead.

PLATFORM DATA PIPELINE FLOW:

User Input → Heuristic Vectorizer → Risk Score Calculation → Threat Lab Forensic Overlay → Visual Remediation



Tactical SOC Dashboard (`Home.tsx`)

Centralized Security Operations Center interface delivering live system telemetry, threat feed, and readiness scoring.

● Telemetry Header & Readiness Meter

- ▶ **Live System Pulse:** Real-time operational indicator: ● SYSTEM OPERATIONAL // ENGINE v1.4.2.
- ▶ **Dynamic Readiness Meter:** Displays the user's defense score (e.g., **88/100** across 12 completed audits), benchmarking resilience against standard social engineering baselines.
- ▶ **Tactical Quick Actions:** One-tap buttons for rapid incident triage, panic reporting, and forensic lab jump.
- ▶ **Zero Generic Aesthetic:** Crafted with high-contrast slate surfaces, hairline borders, and JetBrains Mono technical badges.

⚡ 2×2 Primary Operations Matrix

1. IOC DETECTOR

Live URL & SMS heuristic scanner with instant result overlay.

2. THREAT LAB

Side-by-side cloned site autopsy and interactive radar pins.

3. THREAT CARDS

6 tactical awareness posters with technical forensic breakdown.

4. ATTACK DRILLS

Real-world scenario simulation and timed challenge evaluations.

[LIVE THREAT ADVISORY STREAM]

URGENT ALERT // ACTIVE WAVE

Slick Banking SMS campaign targeting Indian UPI users with fake reward claims.

HEURISTIC FLAGGED // DOMAIN DRIFT

Typosquatted domain `netbanking-hdfc-otp.top` identified and blacklisted.

STATUS // ALL CLEAR

Official OAuth2 identity endpoints verified against global CA roots.

The Threat Lab: Real vs. Fake Analysis (`ThreatLab.tsx`)

A state-of-the-art interactive forensic sandbox allowing users to visually dissect web clones.

Core Capabilities & Workflow

- ▶ **Dual View Modes:** Toggle effortlessly between *Interactive Split View* (side-by-side comparison) and *Single View* with a live Safe/Clone flip switch.
- ▶ **Radar Forensic Hotspots:** Interactive glowing indicator pins superimposed over fraudulent elements (e.g., lookalike domains, fake lock icons, spoofed countdown timers).
- ▶ **Instant Evidence Reveal:** Clicking any hotspot displays an in-depth forensic tooltip explaining the attacker's psychological lure and technical deception mechanism.
- ▶ **Gamified Verification:** Users test their detection intuition and immediately see the autopsy checklist score.

Built-in Real-World Scenarios

TARGET SECTOR	DECEPTIVE ELEMENT	ATTACKER VECTOR
NetBanking Portal	`bank-secure-auth.top`	OTP & Credential Interception
University Portal	`student-portal-campus.example`	Identity & Enrollment Hijack
E-Commerce Mega-Store	Artificial 5-min timer	Credit Card / UPI Theft
Enterprise Cloud (M365)	Unsolicited 2FA prompt	Session Cookie Hijacking

WHY THIS MATTERS: Traditional cybersecurity education tells users "check the lock icon". Attackers now use HTTPS certificates on 83% of phishing sites. PhishGuard's Threat Lab trains users to inspect the **Root Domain**, **Shannon Entropy**, and **Hostname Hierarchy** instead of relying on flawed heuristics.



Dual-Mode Phishing Detector (`Detect.tsx`)

Instant, client-side heuristic inspection engine analyzing suspicious URLs and SMS smishing lures.

🌐 URL Inspection Engine

- ▶ **Hostname Parsing:** Dissects Subdomain, Second-Level Domain, and TLD.
- ▶ **Typosquatting Check:** Detects character swaps and homoglyphs (e.g. `goog1e.com`, `nicrosoft.com`).
- ▶ **Entropy Analysis:** Identifies algorithmic, randomly generated domain strings (DGA).
- ▶ **Port & Protocol Audit:** Flags non-standard web ports and unencrypted HTTP links.

💬 SMS Smishing Engine

- ▶ **Panic Trigger Words:** Scans for "blocked", "immediate action", "KYC update", "suspended".
- ▶ **Masked Redirects:** Automatically isolates shorteners (bit.ly, tinyurl) and raw IPs.
- ▶ **Authority Spoofing:** Identifies fake bank headers (SBI, HDFC, ICICI, PayPal).
- ▶ **Financial Coercion:** Detects fake tax refunds, lottery wins, and lottery claims.

📊 Verdict Classification

0% - 25% RISK: CLEAN

Official domains, trusted TLDs, no suspicious urgency.

26% - 69% RISK: SUSPICIOUS

Subdomain tricks, uncommon TLD, mild urgency cues.

70% - 100% RISK: DANGEROUS

Known phishing indicators, credentials requested, blacklisted strings.

PRIVACY ADVANTAGE: Evaluations execute entirely client-side. The user's confidential SMS text and inspected URLs are never sent to external third-party logging servers.

ZERO DATA LEAKAGE



Tactical Posters & Autopsy Modals (`Posters.tsx`)

Six comprehensive, visual forensic cards delivering technical breakdowns and counter-measures for major attack vectors.

VECTOR 01

Credential Harvesting

Weaponized Microsoft 365 / Google Workspace login clones designed to capture corporate SSO credentials and session cookies.

HIGH SEVERITY

VECTOR 02

Banking Smishing

Fraudulent SMS alerting of sudden account freeze or PAN/KYC invalidation, redirecting victims to cloned OTP verification forms.

FINANCIAL LOSS

VECTOR 03

QR Code Phishing (Quishing)

Counterfeit QR stickers pasted over parking meters, restaurant menus, or phishing emails to bypass secure email gateways.

STEALTH VECTOR

VECTOR 04

Parcel Delivery Scam

Fake courier alerts (FedEx, DHL, India Post) demanding a nominal fee (\$1.50) to release a delivery, harvesting full debit card details.

MASS FRAUD

VECTOR 05

Executive Impersonation

Whaling and CEO fraud messages sent to finance personnel requesting urgent, confidential wire transfers or gift cards.

ENTERPRISE RISK

VECTOR 06

Ransomware Attachment

Weaponized macro-enabled invoices (.docm, .vbs, .iso) masquerading as vendor receipts to deliver Cobalt Strike or LockBit payloads.

CRITICAL IMPACT

TECHNICAL AUTOPSY MODAL: Clicking any card opens a full forensic breakdown modal featuring Indicators of Compromise (IOCs), Attack Lifecycle Diagram, and an actionable Mitigation Checklist for immediate defensive action.



Cybersecurity Academy & Drill Engine (`Learn.tsx`, `Quiz.tsx`)

Structured multi-tier educational path backed by interactive scenario drills and dynamic quiz evaluations.

8-Stage Progressive Curriculum

- ▶ **Stage 01 // Fundamentals:** What defines modern phishing and why traditional spam filters miss 9% of attacks.
- ▶ **Stage 02 // Attack Vectors:** In-depth taxonomy of Spear Phishing, Smishing, Vishing, and Quishing.
- ▶ **Stage 03 // Clone Mechanics:** Reverse-proxy kits (Modlishka, Evilginx) and how they bypass standard 2FA.
- ▶ **Stage 04 // URL Forensic Dissection:** Deep breakdown of Protocols, Subdomains, Root Domains, Paths, and Query Strings.
- ▶ **Stage 05 // Visual Red Flags:** Identifying mismatched logos, broken CSS, and fake SSL security badges.
- ▶ **Stage 06 // Response Protocols:** Clear playbooks for immediate password revocation and bank notifications.
- ▶ **Stage 07 // Practical Drills:** Hands-on challenge evaluating real vs simulated attack strings.
- ▶ **Stage 08 // Threat Lab Capstone:** Graduation into live sandbox clone analysis.

Dynamic Quiz & Readiness Evaluation

REALISTIC SCENARIO EVALUATIONS

Users are presented with realistic scenarios (e.g., student enrollment portal, corporate password reset) and must classify them as *SAFE* or *SUSPICIOUS*.

- ▶ **Instant Technical Rationale:** Detailed forensic explanations provided for both correct and incorrect choices.
- ▶ **Dynamic Score Calculation:** Real-time accuracy tracking directly updates the global SOC Readiness Meter.
- ▶ **Adaptive Difficulty:** From obvious spelling errors to sophisticated homoglyph attacks.
- ▶ **Certification Readiness:** Equips trainees for corporate security audits and phishing readiness compliance.



Enterprise Posture, Telemetry & Campaigns

Equipping teams and organizations with progress telemetry, simulation drills, and security hygiene guidelines.

Digital Posture (`Safety.tsx`)

- ▶ **Hardware Keys (FIDO2):** Guide to implementing phishing-resistant YubiKeys.
- ▶ **Authenticator vs SMS:** Phasing out vulnerable SMS OTPs in favor of TOTP apps.
- ▶ **Password Managers:** Enforcing auto-fill verification to thwart fake domains.
- ▶ **Breach Auditing:** Integration guidelines for checking compromised credentials.

Progress Telemetry (`Progress.tsx`)

- ▶ **Audit History:** Chronological log of all simulated threat evaluations.
- ▶ **Skill Badges:** Unlockable credentials (e.g. *SOC Sentinel*, *Domain Auditor*, *Heuristics Master*).
- ▶ **Streak Counter:** Daily engagement tracking to maintain high security alertness.
- ▶ **Readiness History:** Visual graph of resilience improvements over time.

Attack Simulator (`Campaign.tsx`)

- ▶ **Simulated Campaigns:** Configure mock phishing scenarios for organizations.
- ▶ **Template Library:** IT alerts, urgent HR notifications, delivery updates.
- ▶ **Click-Through Analytics:** Track vulnerability rates across departments.
- ▶ **Automated Retraining:** Route compromised employees directly to remedial modules.

ORGANIZATIONAL COMPLIANCE:

Aligned with **ISO 27001 (A.7.2.2 Information Security Awareness)** and **NIST CSF (PR.AT Awareness and Training)** standards.



Universal Multi-Platform Deployment

Delivering native mobile performance and instant public accessibility across Android, iOS, and Web.

NATIVE ANDROID

Android APK Package

- ▶ Built using **Capacitor 6** native wrapper.
- ▶ Full 5-tier adaptive mipmap icons (mdpi, hdpi, xhdpi, xxhdpi, xxxhdpi).
- ▶ Custom Android splash screen & dark system theme integration.
- ▶ Completely functional offline without external server dependencies.

APPLE iOS

iOS Safari PWA

- ▶ Zero App Store installation barrier or signing certificates.
- ▶ One-tap installation: **Safari → Share → Add to Home Screen**.
- ▶ Launches full-screen in standalone mode without browser chrome.
- ▶ High-resolution iOS touch icons and custom theme color (`#0A0E17`).

GLOBAL WEB

Public Web SOC

- ▶ Live at: <https://shlokt05.github.io/phishguard/>
- ▶ Hosted on GitHub Pages with global Fastly edge acceleration.
- ▶ Auto-renewing TLS 1.3 HTTPS certificate.
- ▶ Automatic deployment pipeline via GitHub Actions on every push.

PROJECT REPOSITORY & LIVE LINKS

GitHub: <https://github.com/shlokt05/phishguard> | Live Web: <https://shlokt05.github.io/phishguard/>

STATUS: PRODUCTION READY